4 February 1975
SUPERSEDING
MIL-M-5388A
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MILITARY SPECIFICATION

MACHMETER; PITOT STATIC OPERATED, RANGE 0.5 TO 1.5, TYPE A-2B

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of 2-3/4 inch round dial, pitot static-operated machmeter, designated Type A-2B.

2. APPLICABLE DOCUMENTS

*2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)
PPP-B-636	Box, Fiberboard

Military

MIL-P-116	Preservation-Packaging, Methods Of
MIL-D-1000	Drawings, Engineering and Associated Lists
MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys
MIL-P-7105	Pipe Threads, Taper, Aeronautical National Form, Symbol ANPT
MIL-S-7742	Screw Threads; Standard, Aeronautical
MIL-A-8625	Anodic-Coatings, for Aluminum and Aluminum Alloys
MIL-L-25142	Luminescent Material Fluorescent

FSC: 6610

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STANDARDS

Military

MIL-STD-129	Marking of Shipments
MIL-STD-130	Identification Marking of U S Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for Selection Of
MIL-STD-794	Parts and Equipment Procedures for Packaging and Packing Of
MIL-STD-810	Environmental, Test Methods
MS28105	Cover Glass Air Force Instrument Dial
MS33558	Numerals and Letters-Aircraft Instrument Dial, Standard Form Of
MS33585	Pointers, Dial, Standard Design of Aircraft Instruments
MS33638	Cases, Instruments, Flange Mounted Aircraft

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3 REQUIREMENTS

*3.1 Qualification. Machmeters furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.2 and 6.3).

3.2 Materials.

3.2.1 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the machmeter, except where magnetic materials are essential.

3.2.2 Metals. Metals shall be of the corrosion-resistant type or shall be suitably protected to resist corrosion during normal service life.

3.2.3 Protective treatment. When materials are used in the construction of the machmeter that are subject to corrosion in salt air or other atmospheric conditions likely to occur during service usage, they shall be protected against such corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of atmospheric conditions shall be avoided.

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*3.2.4 Selection of materials. Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with MIL-STD-143, except as provided in the following paragraph.

*3.2.4.1 Standard parts. Standard parts (MS and AN) shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc, may be used, provided they possess suitable properties and are replaceable by the standard parts (MS or AN) without alteration, and provided the corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings. In the event there is no suitable corresponding standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

3.3 Design. The machmeter shall be designed to indicate the ratio of the speed of the airplane to the speed of sound at varying altitudes and temperatures.

3.3.1 Application of pressure. The pitot pressure shall be applied to the inside of a diaphragm capsule and the static pressure to the interior of the case.

3.4 Construction. The machmeter shall be so constructed that no parts will work loose in service. The machmeter shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.4.1 Pivots, bearings, and gears shall neither bind or shake, and shall be as near frictionless as practicable.

3.4.2 The machmeter shall be so constructed that adjustments and repairs can be easily made by the personnel of operating units and overhaul bases, and shall require a minimum quantity of special tools and fixtures.

3.4.3 Case. The machmeter case shall consist of a body and bezel ring and shall be in accordance with Figure 1.

3.4.3.1 Body. The body shall be made either of nonferrous low-density metal or of synthetic material, uniform in texture with a smooth surface, and entirely covered with a durable dull black finish. Synthetic material shall be of the thermal-setting type, composed of a suitable filler and phenol condensation product binder.

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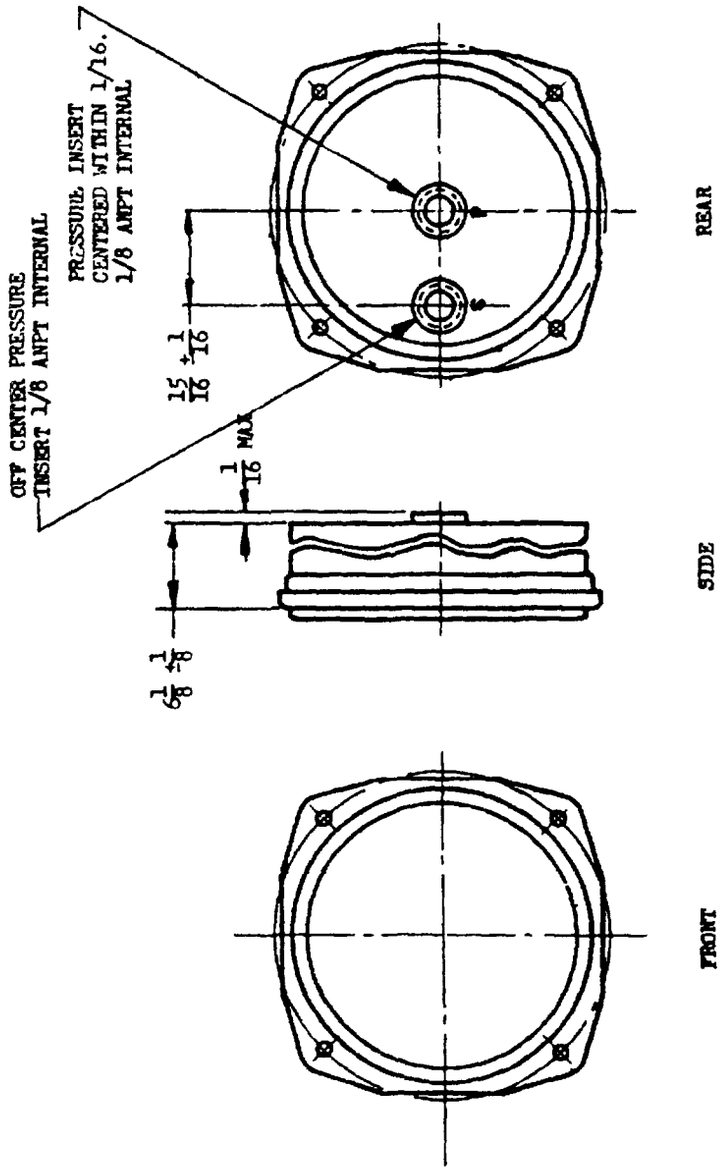


FIGURE 1. Case

DIMENSIONS ARE IN INCHES.

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3.4.3.2 Reinforcement. The inside of the case around each connection shall be of sufficient strength to prevent damage to the case when the connections are tightened during installation of the machmeter.

3.4.3.3 Bezel ring. The bezel ring shall be made of nonferrous, low-density metal, and shall have a durable dull black finish. The ring shall be held in place by means of screws properly secured by lock washers or similar devices of approved design. Provision shall be made for replacement of the cover glass by removal of the bezel ring.

3.4.3.4 The back of the case, adjacent to the connections, shall be marked as follows:

"P" Pressure connection
"S" Static connection

3.4.3.4.1 The following shall be marked on the back of the case:

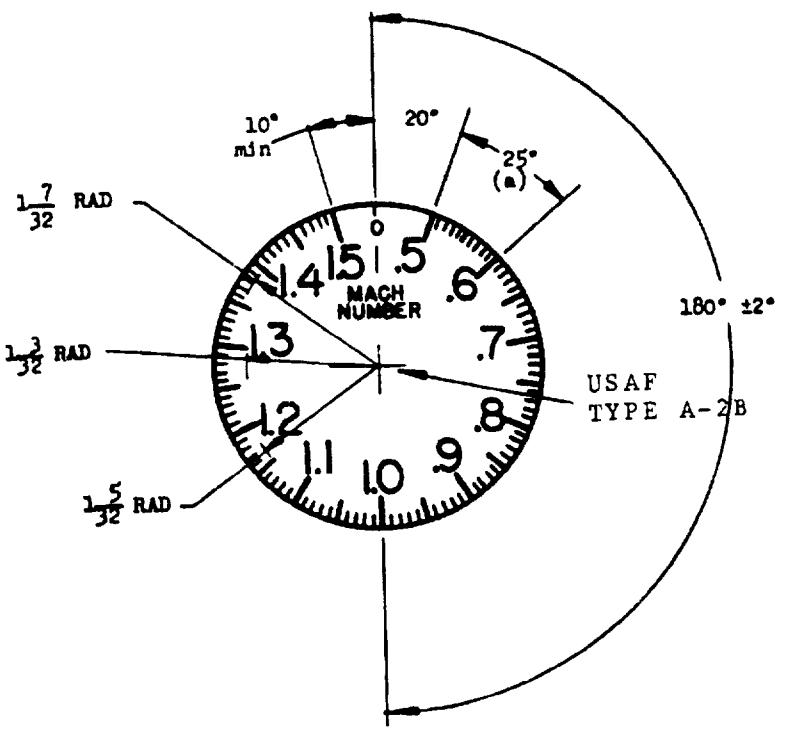
CAUTION: DO NOT BLOW IN TUBES

3.4.3.5 Plug. The static and pitot pressure connections shall be provided with vented plugs of acceptable material and design. Each plug shall incorporate a vent consisting of a hole 1/16 ± 1/64 inch in diameter.

*3.4.4 Cover glass. The cover glass shall be in accordance with MS28105 and shall be properly sealed, provision being made to prevent fogging on the inside of the glass under operating conditions. The distance between the cover glass and the dial shall be the minimum practicable and shall not exceed 0.187 inch.

*3.4.5 Machmeter dial. The machmeter dial shall conform to Figure 2. The style and proportions of numerals and letters placed on the dial shall conform to MS33558. Numerals shall distinctly indicate the graduation to which each applies. Three sample dials shall be submitted to the procuring activity at the beginning of each contract or order for approval. No changes in dial composition, material, or finish shall be made during a contract or order without submitting a like number of samples for approval by the procuring activity.

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(a) NUMERALS FROM .6 TO 1.5 SHALL BE 33° ±5° APART.

DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES:
-ANGLES- ±5° -FRACTIONS- ±1/64

FIGURE 2. Dial

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*3.4.5.1 Fluorescent-luminescent material. The following markings shall be finished in fluorescent-luminescent materials in accordance with Specification MIL-L-25142, Type I or III, as applicable. The dimensions of the markings shall be as follows:

	Height or length Inch ± 1/64	Width of line or graduation Inch ± 0.005
Numerals, except zero	5/16	0.031
Graduations corresponding to numerals, except zero graduation	Figure 2	0.031
Graduations corresponding to 0.55, 0.65, 0.75, 0.85, 0.95, 1.05, 1.15, 1.25, 1.35, and 1.45	Figure 2	0.031
Zero numeral	1/8	0.016
Zero graduation	1/8	0.016
Minor graduations	Figure 2	0.016
Lettering "MACH NUMBER"	1/8	0.016

*3.4.5.2 Durable dull black. The markings for type designation shall be permanently and legibly marked on the dial. These markings, and all other markings not otherwise specified, shall be finished in durable dull black. The height of the markings shall be 3/64 ± 1/64 inch.

3.4.5.3 Visibility of dial. The pointer, numerals, at least 1/16 inch of the shortest graduations, and other specified markings on the dial shall be visible from any point within the frustum of a cone whose side makes an angle of 30 degrees with a perpendicular to the dial, and whose small diameter is the aperture of the machmeter case.

*3.4.5.4 Pointer. The pointer shall be in accordance with MS33585, except that the shaded portion of the pointer shall be finished in fluorescent-luminescent material, in accordance with Specification MIL-L-25142, Type I or III, as applicable.

3.4.5.5 Limitation of pointer movement. The pointer movement shall be limited by suitable stops in the mechanism in such a way that the pointer will not be permitted to rotate more than 355 degrees of arc clockwise from the zero graduation when the machmeter is submitted to the overpressure tests. Suitable stops shall also be incorporated in the machmeter mechanism so as to limit the counterclockwise rotation of the pointer from the zero graduation to a maximum of 1/16 inch. The design of the stops shall be subject to the approval of the procuring activity.

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*3.5 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of Specification MIL-D-1000.

3.6 Weight. The weight of the machmeter completely assembled, less mounting screws, shall not exceed 1.9 pounds.

3.7 Finishes and protective coatings.

3.7.1 Aluminum-alloy parts. Aluminum-alloy parts shall be covered with an anodic film conforming to Specification MIL-A-8625, except as follows:

3.7.1.1 Dials, small holes, and case inserts need not be anodized

3.7.1.2 Aluminum-alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with Specification MIL-C-5541.

3.7.1.3 Where the primary treatment is to afford a suitable paint base, chemical treatments in accordance with Specification MIL-C-5541 may be used in lieu of anodizing.

3.7.1.4 Castings containing nonaluminum alloy integral inserts may be treated with a chemical film in accordance with Specification MIL-C-5541 in lieu of anodizing.

3.7.1.5 When abrasion resistance is a factor, chemical film in accordance with Specification MIL-C-5541 shall not be used in lieu of anodizing.

3.7.2 Steel parts. Steel parts shall be cadmium plated, where practicable, in accordance with Specification QQ-P-416, Type II or III, as applicable, and of a class that is adequate to achieve the degree of protection required.

3.8 Threads.

3.8.1 Screw. Unless otherwise specified, the threads of all machine screws, 0.060 inch or larger in diameter, shall conform to Specification MIL-S-7742.

3.8.2 Pipe. Pipe threads shall conform to Specification MIL-P-7105.

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3.9 Performance. The machmeter shall be capable of meeting the requirements specified in Section 4, when subjected to the applicable tests.

3.10 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with Standard MIL-STD-130.

3.10.1 Use of AN or MIL designations. AN or MIL designations shall not be applied to a product, except for qualification test samples nor referred to in correspondence, until notice of approval has been received from the activity responsible for qualification, or from the Aeronautical Standards Group.

3.11 Installation.

3.11.1 Mounting screws. Mounting screws shall be No. 6-32 NC-2 round brass machine screws having a durable dull black finish. The length shall be sufficient to permit mounting on a panel 3/16 inch thick.

3.11.1.1 Envelope. An envelope containing the mounting screws shall be furnished with each machmeter The following information shall be printed on the face of the envelope:

IMPORTANT
THIS ENVELOPE CONTAINS
MOUNTING SCREWS

3.12 Workmanship. The machmeter, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, marking of parts and assemblies, and freedom of parts from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

*4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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*4.2 Classification of tests. The inspection and testing of machmeters shall be classified as follows:

(a) Qualification Inspection: Qualification inspections are those tests accomplished on samples submitted for qualification as a satisfactory product.

(b) Quality Conformance Inspection: Quality conformance inspections are those tests accomplished on machmeters manufactured and submitted for acceptance under contract.

4.3 Test conditions.

4.3.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of test are not specified definitely, it shall be understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately +25°C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.3.2 Attitude. Unless otherwise specified, the machmeter shall be tested in normal operating position.

4.3.3 Tapping. Unless otherwise specified, the machmeter shall be gently tapped or vibrated before a test reading is taken.

*4.4 Qualification inspection.

*4.4.1 Prior qualification. Unless otherwise specified by the procuring activity, machmeters which have not previously passed a qualification inspection, or which have passed the qualification inspections and have been modified in any manner, shall satisfactorily pass a qualification inspection prior to the acceptance of any machmeter.

*4.4.2 Sampling instructions. Qualification inspection samples shall consist of two machmeters. Samples shall be identified as required and forwarded to the activity responsible for qualification, designated in the letter of authorization from the qualifying activity (see 6.3).

*4.4.3 Tests. The qualification inspections of machmeters shall consist of all the tests specified under quality conformance inspections and, in addition, the following tests subject to the test conditions specified herein.

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*4.4.3.1 Environmental tests. The environmental tests shall be conducted in accordance with Standard MIL-STD-810. The methods and results of tests shall be as specified herein

*4.4.3.1.1 Low temperature exposure: Method 502. While still at -54° ± 2°C, the machmeter shall be operated throughout its range without exhibiting sticking or erratic characteristics, and shall then be subjected to and meet the requirements of the low temperature operation test. There shall be no damage which would adversely affect subsequent operation.

*4.4.3.1.2 High temperature exposure: Method 501, Procedure I. The machmeter shall operate without indications of sticking or erratic action. After the chamber has been returned to room temperature and the machmeter has saturated at the room temperature for minimum period of 4 hours, it shall meet the individual test requirements specified herein. There shall be no damage which would adversely affect subsequent operation.

*4.4.3.1.3 Humidity: Method 507, Procedure II. After the machmeter has returned to room temperature, it shall be subjected to the scale error at room temperature test, as specified herein. The scale errors shall not exceed the tolerances specified for scale error at room temperature. The machmeter shall be examined and there shall be no evidence of corrosion or deterioration which will affect subsequent operation.

*4.4.3.1.4 Vibration failure: Method 514, Procedure I, Curve C. The test shall be conducted while the machmeter is subjected to the altitude and Mach Number specified in Table I. After vibration, the machmeter shall be tested as specified for scale error at room temperature, except only at the points marked with an asterisk in Table II, beginning with the first asterisk. The errors at the test points shall not exceed the tolerance specified in Table II by more than the tolerance specified in Table I. In addition, no looseness in the mechanism nor damage to any part of the machmeter shall result from the vibration. This test shall be conducted on one of the machmeters at the conclusion of the other tests, exept fungus resistance.

*4.4.3.1.5 Fungus resistance: Method 508. The test shall be conducted at the conclusion of other special or qualification tests by disassembling one of the machmeters to expose the internal mechanism to the fungus spores. The machmeter selected shall not have previously been subjected to the salt spray test. At the end of the test period, the machmeter shall be carefully examined to ascertain that no fungus growth has occurred.

*4.4.3.1.6 Salt spray: Method 509. At the end of the 50-hour period, the second machmeter shall be used and subjected to the individual tests specified herein.

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TABLE I

Allowable tolerances for

Tests	Tolerance - Mach No.
Friction	0.02
Position error	0.02
Vibration error	
Altitude 0 ft	
At 0.50 Mach	0.01
At 1.00 Mach	0.01
Altitude 20,000 ft	
At 0.50 Mach	0.01
At 1.00 Mach	0.01
Vibration failure	
Test altitude 20,000 ft	
Test Mach No. 0.70	0.02
Seasoning	
Test altitude 10,000 ft	0.02
Low temperature	0.02
High temperature	0.02
Overpressure	0.02
Pitot pressure connection, 80 in. Hg	
Static pressure connection, 6 in. of water	
Scale error	0.02
High altitude	0.02

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TABLE II

Scale error tolerances

Pressure altitude (ft) P_s (in.Hg)	Mach No.	ΔP (in.Hg)	Pressure altitude (ft) P_s (in.Hg)	Mach No.	ΔP (in Hg)
0	0.50	5.571 *	15,000	0.50	3.143 *
29.9212	0.60	8.243	16.8816	0.60	4.651
	0.70	11.585 *		0.70	6.536 *
	0.80	15.691		0 80	8.853
	0.90	20.682 *		0.90	11.668 *
	1.00	26.717		1.00	15.074
	1.10	33.889 *		1.10	19.120 *
	1.20	42.111		1.20	23.759
	1.30	51.279 *		1.30	28.932 *
	1.40	61.329		1.40	34.602
	1.50	72.212 *		1.50	40.742 *
5,000	0.50	14.635 *	20,000	0.50	2.559 *
24.8940	0.60	6.858	13.7453	0.60	3.787
	0.70	9.639 *		0.70	5.322 *
	0.80	13.054		0.80	7.208
	0.90	17.207 *		0.90	9.501 *
	1.00	22.228		1.00	12.273
	1.10	28.195 *		1.10	15.568 *
	1.20	35.036		1.20	19.345
	1.30	42.663 *		1.30	23.557 *
	1.40	51.025		1.40	28.174
	1.50	60.079 *		1.50	33.173 *
10,000	0.50	3.831 *	35,000	0.50	1.310 *
20.5736	0.60	5.668	7.0356	0.60	1.938
	0.70	7.966 *		0.70	2 724 *
	0.80	10.789		0.80	3.689
	0.90	14.220 *		0.90	4.863 *
	1.00	18.370		1.00	6.282
	1.10	23.302 *		1.10	7.968 *
	1.20	28.955		1.20	9.902
	1.30	35.259 *		1.30	12.058 *
	1.40	42.170		1.40	14.421
	1.50	49.652 *		1.50	16.980 *

* Test points

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TABLE II (Continued)

Scale error tolerances

Pressure altitude (ft) P_s (in. Hg)	Mach No.	ΔP (in Hg)	Pressure altitude (ft) P_s (in Hg)	Mach No.	ΔP (in Hg)
50,000 3.4370					
	0 70	1.331 *			
	0.80	1.802			
	0.90	2.376 *			
	1.00	3.069			
	1.10	3.893 *			
	1.20	4.837			
	1.30	5.890 *			
	1.40	7.045			
	1.50	8.295 *			

Table II is calculated in accordance with the following equations:

$$\frac{P_p}{P_s} = \left(\frac{M^2}{5} + 1\right)^{7/2} \quad \text{when } M \leq 1$$

$$\frac{P_p}{P_s} = \frac{166.9215\ M^7}{(7M^2 - 1)^{5/2}} \quad \text{when } M \geq 1$$

where M = Mach Number
P_p = Pressure in Pitot Tube
P_s = Static Pressure

$\Delta P = P_p - P_s$

* Test points

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4.5 Quality conformance inspection. Quality conformance inspection shall consist of individual tests and sampling tests.

4.5.1 Individual tests. Each machmeter shall be subjected to the following tests.

4.5.1.1 Examination of product. The machmeter shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, proper operation, and performance.

4.5.1.2 Leakage.

4.5.1.2.1 Case leakage. The pitot and static pressure connections of the machmeter shall be joined with a "Y" connection and shall be connected together to the same mercury manometer and sources of suction and pressure. A suction of 15 inches Hg shall be slowly applied to the pitot and static pressure connections. With sources disconnected during a period of 10 seconds, the difference between the mercury levels in the manometer shall not change by more than 0.2 inch.

4.5.1.2.2 Diaphragm capsule leakage. The static connection shall be opened to the atmosphere and a pressure sufficient to produce approximately full-scale deflection of the pointer shall be applied to the pitot connection of the machmeter at which point the connecting tubing shall be pinched off or otherwise completely sealed. During a period of 1 minute, the poin pointer shall hot change its position by more than 0.02 Mach Number.

4.5.1.3 Scale error at room temperature. The machmeter shall be tested for scale errors at the points of the scale indicated in Table II. During this test, the static pressure connection shall be suitably connected to an accurate barometer, a source of air pressure and suction, and to the suction side of water and mercury manometers.

4.5.1.3.1 The pitot pressure connection shall be connected to a source of air pressure and suction and to the pressure side of the water and mercury manometers.

4.5.1.3.2 The static pressure shall be adjusted to the standard value (29.92 inches Hg) corresponding to zero feet altitude, sea level standard as specified in Table II.

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4.5.1.3.3 The static pressure shall be held constant, and the scale error tests accomplished by subjecting the machmeter to the differential pressures specified to produce the readings, first with pressures increasing, then with pressures decreasing. With pressures increasing, the pressure shall be brought up to, but shall not exceed the pressure specified to obtain the desired reading; and with pressures decreasing, the pressure shall be brought down to, but shall not fall below the pressure specified, to obtain the desired reading. The error at the test points shall not exceed 0.008 Mach Number for the 57 test points, 0.010 Mach Number for 13 test points, and 0.015 Mach Number for 7 test points. This procedure shall be repeated for each of the altitudes specified in Table II.

4.5.2 Sampling tests. Five machmeters shall be selected at random from each lot of 500 or fraction thereof on the order and subjected to the following sampling tests. These tests shall be in addition to the individual tests specified herein.

4.5.2.1 Weight. The machmeter shall be weighed completely assembled, and without connection fittings and mounting screws. The weight of the machmeter shall not exceed 1.9 pounds.

4.5.2.2 Magnetic property. The magnetic property of the machmeter shall be determined in terms of the deflection of a free magnet, approximately 1-1/2 inches long, in a magnetic field with a horizontal intensity of 0.18 $\pm$ 0.01 gauss, when the machmeter is held in various positions on an East-West line 5 inches from the center of the magnet. An aircraft compass with compensating magnets removed therefrom may be used as the free magnet for this test. The maximum deflection of the magnet shall not exceed 1 degree.

4.5.2.3 Friction. The machmeter shall be tested for friction at every other point in Table II beginning with the first. The pressure shall be increased to bring the pointer approximately to the desired reading, and then held constant while two readings are taken, the first before, and the second after the machmeter is tapped. The difference between any two such readings shall not exceed the tolerance specified in Table I. The pointer shall move smoothly, while the pressure is varied uniformly, without vibration of the machmeter. The test may be combined with the test for scale errors.

4.5.2.4 Position error. The machmeter reading shall be observed while the machmeter is held in any desired position and tapped. This reading shall not differ from the reading, when the machmeter is held in any other position, by more than the tolerances specified in Table I. This test shall be accomplished at every other one of the points in Table II beginning with the first.

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*4.5.2.5 Low temperature operation. The machmeter shall be subjected to a temperature of -54° ± 2°C for a period of 4 hours. At the end of this period and while still at the low temperature, the indicator shall be tested for scale error as specified in 4.5.1.3. The errors at the test points shall not differ from the corresponding errors in the scale error at room temperature test by more than the tolerances specified in Table I.

*4.5.2.6 High temperature operation. The machmeter shall be placed in a chamber and maintained at a temperature of 71° ± 2°C for a period of 4 hours. While the machmeter is still at this temperature, it shall be tested for scale error as specified in 4.5.1.3. The errors at the test points shall not differ from the corresponding errors in the scale error at room temperature test by more than the tolerances specified in Table I.

4.5.2.7 Vibration error. The machmeter shall be so mounted on the vibration stand that a line joining the center of the two lower mounting holes is in a horizontal plane, and the plane of the mounting lugs in a vertical position. The machmeter shall be subjected to the altitudes and Mach Numbers specified in Table I and to vibration with an amplitude between 0.003 and 0.005 inch at frequencies of from 300 to 3,000 cpm.

4.5.2.7.1 Pointer vibration. The maximum amplitude of the pointer vibration and the pointer variation shall not exceed 0.01 Mach Number.

4.5.2.7.2 Accuracy. The reading shall not differ at any time by more than 0.01 Mach Number from the reading obtained when the indicator is at rest.

4.5.2.8 Overpressure.

4.5.2.8.1 The pressure specified in Table I shall be applied to the pitot connection of the machmeter for a period of 10 minutes.

4.5.2.8.2 The pressure specified in Table I shall be applied to the static connection of the machmeter for a period of 10 minutes.

4.5.2.8.3 Not less than 5 minutes following the application of overpressure, the machmeter shall be tested for scale error as specified in the scale error at room temperature test. The errors at the test points shall not differ from the corresponding errors in the scale error at room temperature test by more than the tolerance specified in Table I.

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4.5.2.9 Seasoning. The machmeter shall be subjected to 1,000 applications of a differential pressure sufficient to produce approximately full-scale deflection. These applications shall be made at a rate not greater than 6 per minute. Not less than 1 hour following this test, the machmeter shall be tested for scale errors as specified for scale error at room temperature test at the test altitude specified in Table I. The errors at the test points shall not differ from the corresponding errors in the scale error at room temperature test by more than the tolerance specified in Table I.

4.5.2.10 High altitude

4.5.2.10.1 The static pressure connection of the machmeter shall be connected to a source of vacuum through a control valve and to a mercury barometer. The pitot pressure connection shall be connected to a source of vacuum through a separate control valve.

4.5.2.10.2 Vacuum shall slowly be applied to the static pressure connection and when the pointer has reached 1.2 Mach Number vacuum shall also be applied to the pitot pressure connection so as to maintain the indication between 1.2 and 1.5 Mach Number until a pressure altitude of 80,000 feet (0.820 inch Hg absolute) is reached in the static line. Both lines shall then be sealed to maintain the pressure for a period of 5 minutes, after which the Mach Number indication shall be reduced to and maintained between 0.5 and 1.0 Mach Number, while the static pressure is slowly increased to atmospheric. As the static pressure approaches atmospheric pressure the Mach Number shall be allowed to reach zero.

4.5.2.10.3 Not less than 15 minutes or more than 30 minutes following the reduction of altitude to zero, the machmeter shall be tested for scale error as specified in the scale error at room temperature test. The errors at the test points shall not differ from the corresponding errors in the scale error at room temperature test by more than the tolerance specified in Table I.

4.5.2.11 Pointer zero position error. The zero position of the pointer, with no pressure applied to the indicator, shall be '0' within $\pm$ 1/16 inch.

4.5.3 Rejection and retest. When tests are specified on a quantity of machmeters that are selected as representative of a certain lot, and one or more of this number fails to meet the requirements, additional machmeters of the lot represented shall be tested immediately to determine the cause of failure. Individual performance tests shall not be interrupted, unless the defect is of such a nature that it will seriously affect the per-

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formance or safe use of the machmeter. Rejected machmeters shall not be resubmitted for inspection without furnishing full particulars concerning previous rejection and measures taken to overcome the defects.

4.6 All parts, specimens, or assemblies destroyed in making tests required by this specification or drawings, to determine compliance with the specification or drawings, shall be in addition to the quantity specified in the contract or purchase order and shall be furnished without increasing the cost of the contract or order.

4.7 Inspection and test. Tests for methods of preservation and packaging will be accomplished in accordance with Section 4 of MIL-P-116 to assure compliance with Section 5 of this specification.

5. PREPARATION FOR DELIVERY

*5.1 Preservation and packaging.

*5.1.1 Level A. (Reference definition in MIL-STD-794) Machmeters shall be preserved and packaged in quantity unit packs of one each in accordance with Method IIC of MIL-P-116. The unit container shall conform to PPP-B-636, Weather-Resistant Class. Cushioning material between bag and unit container to be of a type, density and thickness that will insure shock transmission does not exceed the peak values in Gs established for the item when completed packs are subjected to the rough handling drop tests stipulated by MIL-STD-794.

*5.1.2 Level C. Unless otherwise specified, indicators shall be preserved and packaged in a manner to prevent deterioration and damage during shipment from the supply source to the first receiving activity.

*5.2 Packing.

*5.2.1 Level A. Machmeters packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, Weather-Resistant Class. Insofar as possible shipping containers shall be uniform shape, size, of minimum tare and cube consistant with the protection required.

*5.2.2 Level B. Machmeters packaged as specified in 5.1.1 shall be packed in PPP-B-636, Domestic Class Boxes. Other requirements specified in 5.2.1 shall apply.

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*5.2.3 Level C. Packaging shall be applied which affords adequate protection against physical and mechanical damage during direct domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations and may be the contractor's commercial practice provided the latter meets the requirement of this level.

*5.3 Marking.

5.3.1 Container markings. Unit containers and shipping containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be as follows:

Machmeter, Pitot Static Operated, Range 0.5 to 1.5, Type A-2B

6. NOTES

6.1 Intended use. The Type A-2B machmeter covered by this specification is intended for use in indicating the ratio of the speed of the airplane to the speed of sound at a particular altitude and temperature. The range is 0.5 to 1.5 Mach Number at altitude of zero to 50,000 feet.

6.2 Ordering data. Procurement documents should specify the title, number, and date of this specification.

*6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is Oklahoma City ALC/MMEOT, Tinker AFB, Oklahoma 73145 and information pertaining to qualification of products may be obtained from that activity.

6.4 Definitions.

6.4.1 Amplitude. Whenever the word "amplitude" is specified, it shall mean the extent of motion as measured from one extreme to the opposite extreme.

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*6.5 Marginal notations. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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